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""" Rule-Based Chatbot ----- A simple chatbot that uses pattern matching (regex) and
if-else logic to recognize user intents and respond with predefined answers. Run it with: python
chatbot.py Type 'quit', 'exit', or 'bye' to end the conversation. """ import re import random from
datetime import datetime # ----- # 1. Define
rules as (pattern, [possible responses]) pairs. # Patterns are regexes matched (case-
insensitively) against user input. # {0}, {1}, ... in a response are filled in with regex capture
groups. # ----- RULES = [
(r"\b(hi|hello|hey|greetings)\b", ["Hello there! How can I help you today?", "Hi! What can I do for
you?"]), (r"\bmy name is (\w+)\b", ["Nice to meet you, {0}!", "Hello, {0}! Great to have you here."]),
(r"\bwhat(?:'s|) is your name\b", ["I'm RuleBot, a simple rule-based chatbot.", "You can call me
RuleBot."]), (r"\bhow are you\b", ["I'm just a program, but I'm running smoothly! How about
you?", "Doing great, thanks for asking!"]), (r"\b(what time|current time)\b", ["The current time is
{0}."]), # special-cased below (r"\b(what(?:'s|) is the date|today's date)\b", ["Today's date is
{0}."]), # special-cased below (r"\bhelp\b", ["I can chat about basic topics. Try greeting me, telling
me your " "name, or asking for the time/date."]), (r"\b(thanks|thank you)\b", ["You're welcome!",
"No problem at all!"]), (r"\b(bye|goodbye|see you)\b", ["Goodbye! Have a great day.", "See you
later!"]), ] def get_response(user_input: str) -> str: """Match the user input against the rule set
and return a response.""" text = user_input.strip() for pattern, responses in RULES: match =
re.search(pattern, text, re.IGNORECASE) if match: response = random.choice(responses) #
Special handling for dynamic content if "current time" in pattern or "what time" in pattern: return
response.format(datetime.now().strftime("%H:%M:%S")) if "today's date" in pattern or "date" in
pattern: return response.format(datetime.now().strftime("%Y-%m-%d")) # Fill in any captured
groups (e.g. the user's name) groups = match.groups() if groups: return
response.format(*groups) return response # Fallback response when no rule matches fallback =
[ "I'm not sure I understand. Could you rephrase that?", "Interesting! Tell me more.", "I don't have
a rule for that yet, but I'm listening.", ] return random.choice(fallback) def main(): print("RuleBot:
Hello! I'm a rule-based chatbot. Type 'quit' to exit.\n") while True: try: user_input = input("You: ")
except (EOFError, KeyboardInterrupt): print("\nRuleBot: Goodbye!") break if
user_input.lower().strip() in {"quit", "exit", "bye"}: print("RuleBot: Goodbye! Have a nice day.")
break reply = get_response(user_input) print(f"RuleBot: {reply}") if __name__ == "__main__":
main()

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